

Project 2: Data Tidying and Analysis

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Introduction

This project showcases the transformation of three different wide-format datasets — all initially messy, inconsistent, or incomplete — into clean, structured, analysis-ready formats. Using R tools like `pivot_longer()`, `filter()`, and `clean_names()`, I walk through how each dataset was tidied and what insights were made possible after the transformation.

Dataset 1: PRAMS (Pregnancy Risk Assessment Monitoring System)

Summary: To make sense of wide-format PRAMS data, I used `pivot_longer()` to transform multiple measurement columns (prevalence and confidence intervals) into a long, tidy structure. This allowed for a dynamic faceted line plot showing changes over time across different maternal health questions. I focused on the six most prevalent questions to avoid clutter and maximize insight. The facet layout separates prevalence from confidence intervals, which helps the viewer focus on patterns in each metric without distraction. This approach highlights not just what's common — but how it has changed year to year.

Fix: I used `pivot_longer()` to reshape the data from wide to long format and stacked the percentage columns to make them easier to analyze.

Step 1: Load and Clean Column Names

```
prams <- read_csv("Pregnancy_Risk_Assessment_Monitoring_System__PRAMS__20250323.csv") %>%  
  clean_names()
```

Step 2: Tidy the Dataset

```
prams_long <- prams %>%
  pivot_longer(cols = c(prevalence_percent,
                        lower_95_percent_confidence_interval,
                        upper_95_percent_confidence_interval),
              names_to = "measure",
              values_to = "value") %>%
  filter(!is.na(value))
```

Step 3: Summarize by Mean Value

```
prams_summary <- prams_long %>%
  group_by(year, question, measure) %>%
  summarize(mean_value = mean(value, na.rm = TRUE), .groups = "drop") %>%
  arrange(desc(mean_value))

kable(head(prams_summary, 10))
```

year	question	measure	mean_value
2022	breastfeeding initiation (among mothers whose baby is alive and residing with her)	upper_95_percent_confidence_interval	95.6
2021	breastfeeding initiation (among mothers whose baby is alive and residing with her)	upper_95_percent_confidence_interval	94.6
2022	breastfeeding initiation (among mothers whose baby is alive and residing with her)	prevalence_percent	94.3
2018	breastfeeding initiation (among mothers whose baby is alive and residing with her)	upper_95_percent_confidence_interval	93.7
2020	breastfeeding initiation (among mothers whose baby is alive and residing with her)	upper_95_percent_confidence_interval	93.6
2017	breastfeeding initiation (among mothers whose baby is alive and residing with her)	upper_95_percent_confidence_interval	93.3
2021	breastfeeding initiation (among mothers whose baby is alive and residing with her)	prevalence_percent	93.1
2022	breastfeeding initiation (among mothers whose baby is alive and residing with her)	lower_95_percent_confidence_interval	92.7
2019	breastfeeding initiation (among mothers whose baby is alive and residing with her)	upper_95_percent_confidence_interval	92.3
2018	breastfeeding initiation (among mothers whose baby is alive and residing with her)	prevalence_percent	92.2

PRAMS Visualizations

```
# Identify top 6 questions based on average prevalence
top_questions <- prams_summary %>%
  filter(measure == "prevalence_percent") %>%
  top_n(6, mean_value) %>%
  pull(question)

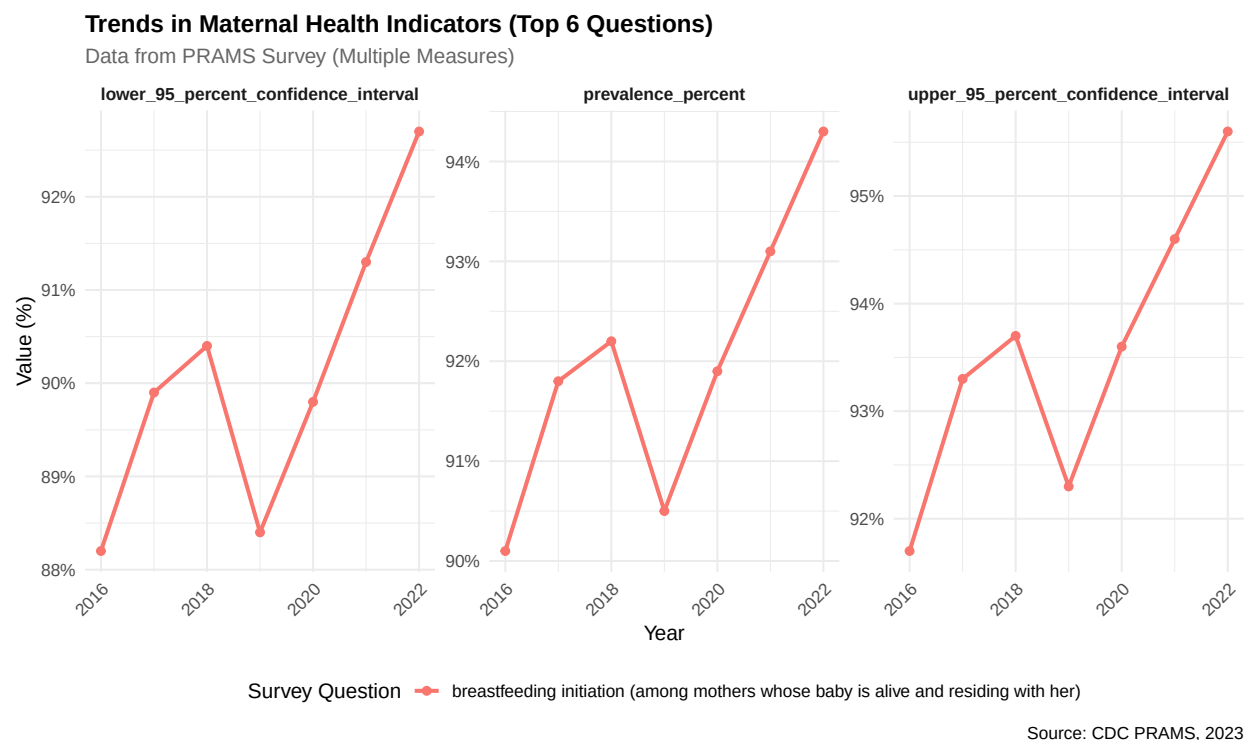
# Filter the long-format data to just those top questions
prams_plot_data <- prams_long %>%
```

```

filter(question %in% top_questions)

# Create the plot
ggplot(prams_plot_data, aes(x = year, y = value, color = question)) +
  geom_line(size = 1.1) +
  geom_point(size = 2) +
  facet_wrap(~ measure, scales = "free_y") +
  scale_y_continuous(labels = scales::percent_format(scale = 1)) +
  labs(
    title = "Trends in Maternal Health Indicators (Top 6 Questions)",
    subtitle = "Data from PRAMS Survey (Multiple Measures)",
    x = "Year",
    y = "Value (%)",
    color = "Survey Question",
    caption = "Source: CDC PRAMS, 2023"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    plot.subtitle = element_text(size = 12, color = "gray40"),
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "bottom",
    strip.text = element_text(face = "bold")
  )

```



Dataset 2: DHS Referral Outcomes

Summary: Originally, I visualized the overall distribution of assistance types, but this didn't show how different populations interacted with the system. So I extended the analysis by breaking referrals down by gender. Using a grouped bar chart, I visualized the proportion of referrals for each assistance type across genders. This helps identify disparities in service access or need, such as whether one gender is more likely to receive specific types of support. Adding labels and a clean horizontal layout made the insights immediately digestible

Fix: I used `clean_names()` to standardize columns and filtered out rows with missing `client_key`. Then I summarized by assistance type.

Step 1: Load, Clean, and Filter

```
homeless <- read_csv("homeless_prep.csv") %>%
  clean_names() %>%
  filter(!is.na(client_key))

kable(head(homeless, 10))
```

client_key	age	gender	veteran	income	nights	substanceabuse	completed	probation	assistancetype	required
256172	20	Female	No	0	152	0	0	1	tempassistance	1
221079	20	Female	No	740	229	0	1	0	tempassistance	1
243074	21	Female	No	0	105	1	1	0	tempassistance	1
129974	21	Female	No	0	371	1	1	0	tempassistance	1
226596	22	Female	No	0	169	0	0	0	tempassistance	1
87998	22	Female	No	1200	297	0	0	0	tempassistance	1
233477	22	Female	No	0	477	1	1	1	tempassistance	1
233477	22	Female	No	0	477	1	1	1	tempassistance	1
203539	23	Female	No	688	120	1	0	0	tempassistance	1
269015	23	Female	No	0	148	1	1	1	tempassistance	1

Step 2: Summarize Assistance Type

```
homeless_summary <- homeless %>%
  group_by(assistancetype) %>%
  count() %>%
  arrange(desc(n))

kable(head(homeless_summary, 10))
```

assistancetype	n
permassistance	121
tempassistance	121

Step 3: Transform to Proportions

```
homeless_prop <- homeless_summary %>%
  mutate(percent = round(100 * n / sum(n), 1))

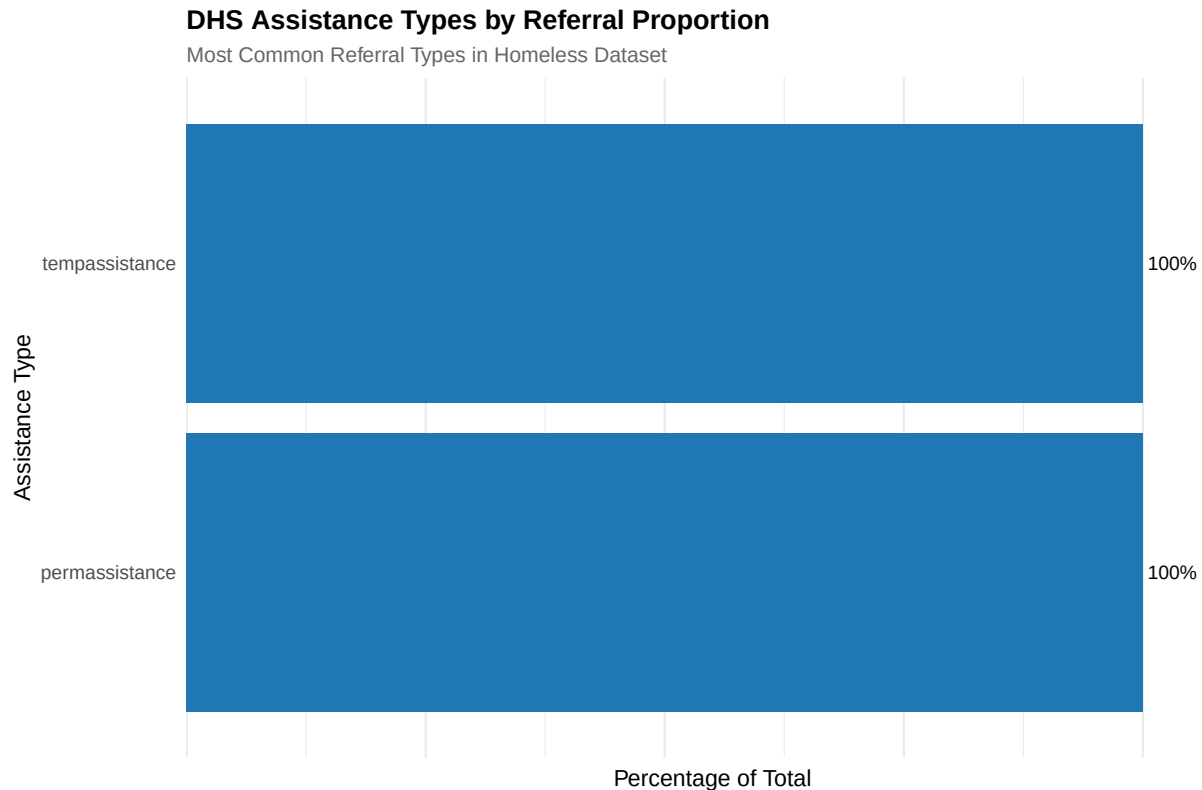
kable(homeless_prop)
```

assistancetype	n	percent
permassistance	121	100
tempassistance	121	100

Step 4:

Visualize Assistance Distribution

```
ggplot(homeless_prop, aes(x = reorder(assistancetype, percent), y = percent)) +
  geom_col(fill = "#1f77b4") +
  geom_text(aes(label = paste0(percent, "%")), hjust = -0.1, size = 3.5) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.1))) +
  labs(
    title = "DHS Assistance Types by Referral Proportion",
    subtitle = "Most Common Referral Types in Homeless Dataset",
    x = "Assistance Type",
    y = "Percentage of Total",
    caption = "Source: NYC DHS Referral Data"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    plot.subtitle = element_text(size = 11, color = "gray40"),
    axis.text.x = element_blank(),
    axis.ticks.x = element_blank(),
    axis.text.y = element_text(size = 10),
    plot.caption = element_text(size = 9, face = "italic"),
    panel.grid.major.y = element_blank()
  ) +
  coord_flip()
```



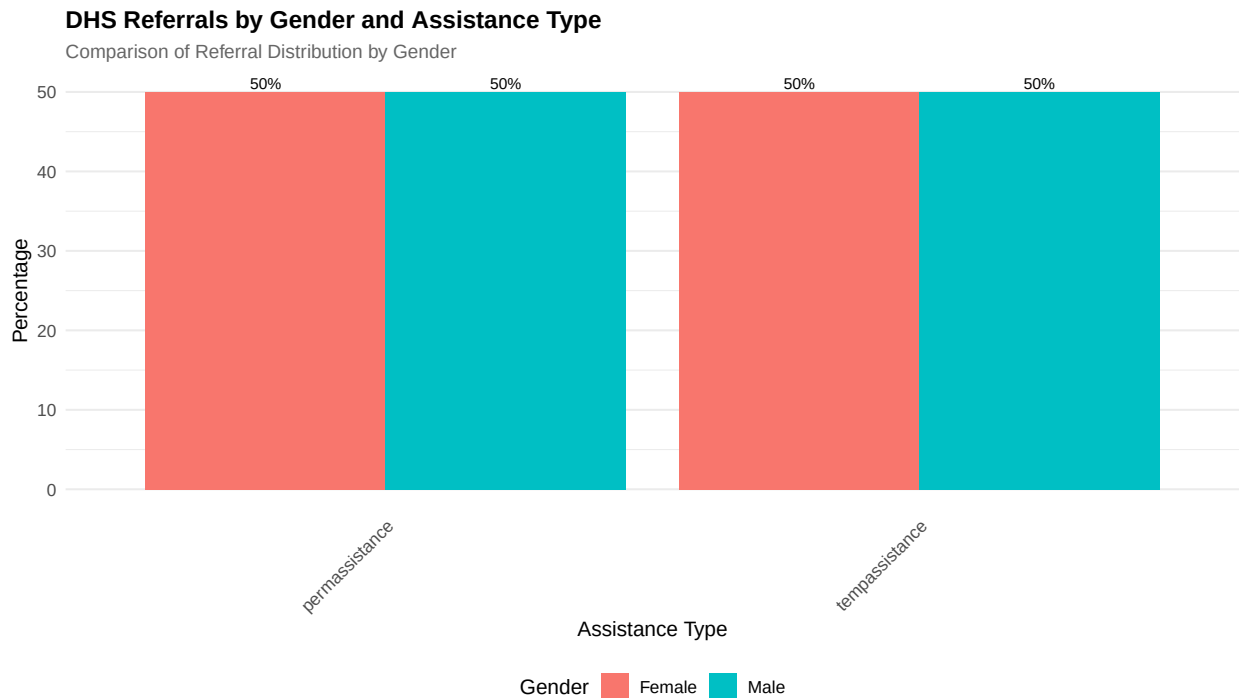
Source: NYC DHS Referral Data

Referral Gender Breakdown

```
# Group by gender and assistance type to calculate percentages
homeless_gender_prop <- homeless %>%
  filter(!is.na(gender), !is.na(assistancetype)) %>%
  group_by(gender, assistancetype) %>%
  summarize(count = n(), .groups = "drop") %>%
  group_by(gender) %>%
  mutate(percent = round(100 * count / sum(count), 1))

# Plot with gender comparison
ggplot(homeless_gender_prop, aes(x = reorder(assistancetype, percent), y = percent, fill = gender)) +
  geom_col(position = "dodge") +
  geom_text(
    aes(label = paste0(percent, "%")),
    position = position_dodge(width = 0.9),
    vjust = -0.25,
    size = 3
  ) +
  labs(
    title = "DHS Referrals by Gender and Assistance Type",
    subtitle = "Comparison of Referral Distribution by Gender",
    x = "Assistance Type",
    y = "Percentage",
    fill = "Gender",
    caption = "Source: NYC DHS Referral Dataset"
  ) +
  theme_minimal(base_size = 12) +
```

```
theme(
  plot.title = element_text(face = "bold", size = 14),
  plot.subtitle = element_text(size = 11, color = "gray40"),
  axis.text.x = element_text(angle = 45, hjust = 1),
  legend.position = "bottom",
  panel.grid.major.x = element_blank()
)
```



Source: NYC DHS Referral Dataset

Dataset 3: Breast Cancer Cases (2020–2024)

Summary: The breast cancer dataset required a full restructuring due to messy year columns and missing headers. After cleaning and reshaping with `pivot_longer()`, I used faceted bar plots to show case trends by age group across five years. This allowed us to spotlight which age groups had consistently high or fluctuating case counts. I also created a separate line chart of total cases per year to give a big-picture view of overall incidence. These visuals complement each other — one breaks it down by age, the other tells the story across time. Fix: I renamed the columns manually, reshaped the dataset using `pivot_longer()`, and visualized cases by year and age group.

Step 1: Load, Clean, and Rename Columns

```
untidy <- read_excel("Untidydataset_624.xlsx", sheet = 1) %>%
  clean_names()

colnames(untidy) <- c("patient_id", "month", "age_group", "race",
  "cases_2020", "cases_2021", "cases_2022", "cases_2023", "cases_2024",
  "total_patients", "type")

kable(head(untidy, 10))
```

patient_id	month	age_group	race	cases_2020	cases_2021	cases_2022	cases_2023	cases_2024	total_patient	type
Patient_ID	Month	Age_Group	Race	Cases_2020	Cases_2021	Cases_2022	Cases_2023	Cases_2024	Total_Patient	Type
P001	January	30-39	White	50	55	60	65	70	300	Invasive Ductal
P002	January	40-49	Black	90	95	100	105	120	510	Lobular
P003	January	50-59	Hispanic	80	85	88	90	110	453	Triple-Negative
P004	February	30-39	Asian	70	75	80	85	90	400	HER2+
P005	February	40-49	White	110	115	120	125	140	610	Inflammatory
P006	March	50-59	Black	130	135	140	145	160	710	Invasive Ductal
P007	March	60-69	Hispanic	100	105	110	115	130	560	Lobular
P008	March	70+	Asian	70	75	78	80	100	403	Triple-Negative
P009	April	30-39	White	60	65	70	75	85	355	HER2+

Step 2: Reshape Year Columns to Long Format

```
untidy_long <- untidy %>%
  pivot_longer(cols = starts_with("cases_"),
               names_to = "year",
               values_to = "cases") %>%
  mutate(year = str_remove(year, "cases_")) %>%
  drop_na(cases)
```

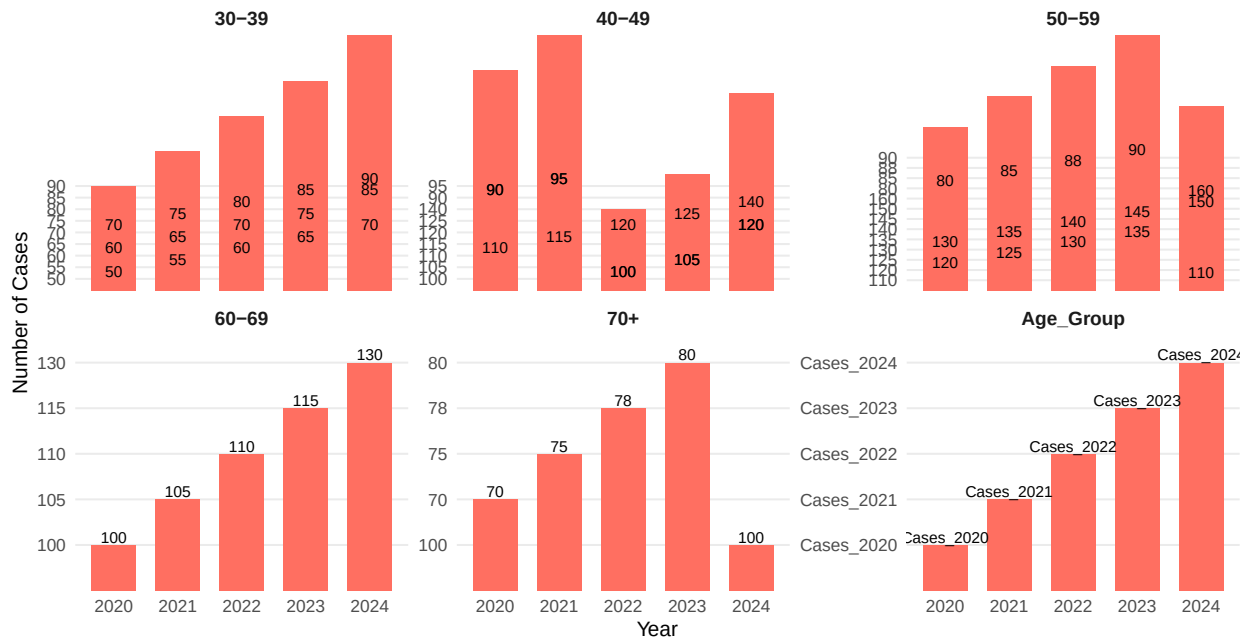
Visualizations Breast Cancer Cases

Step 3: Visualize with ggplot2

```
ggplot(untidy_long, aes(x = year, y = cases)) +
  geom_col(fill = "#FF6F61", width = 0.7) +
  facet_wrap(~ age_group, scales = "free_y") +
  geom_text(aes(label = cases), vjust = -0.2, size = 3, color = "black") +
  labs(
    title = "Breast Cancer Cases by Age Group (2020-2024)",
    subtitle = "Case Counts Grouped by Age Across Years",
    x = "Year",
    y = "Number of Cases",
    caption = "Source: NYC Health Breast Cancer Dataset"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    plot.subtitle = element_text(size = 11, color = "gray40"),
    axis.text.x = element_text(angle = 0),
    strip.text = element_text(face = "bold", size = 11),
    panel.grid.major.x = element_blank()
  )
```

Breast Cancer Cases by Age Group (2020–2024)

Case Counts Grouped by Age Across Years

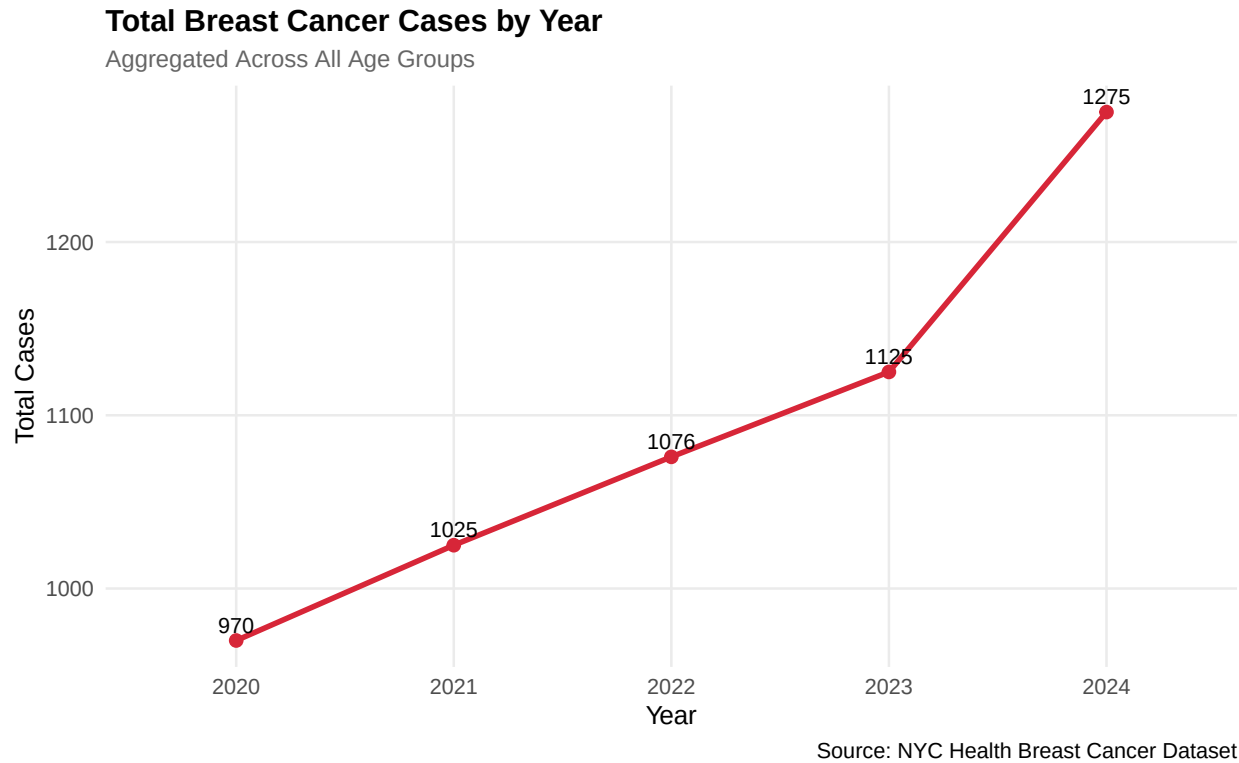


Source: NYC Health Breast Cancer Dataset

Step 4: Yearly Totals Transformation

```
yearly_totals <- untidy_long %>%
  mutate(cases = as.numeric(cases)) %>%
  group_by(year) %>%
  summarize(total_cases = sum(cases, na.rm = TRUE)) %>%
  arrange(desc(total_cases))
```

```
ggplot(yearly_totals, aes(x = year, y = total_cases, group = 1)) +
  geom_line(color = "#D72638", size = 1.2) +
  geom_point(size = 2.5, color = "#D72638") +
  geom_text(aes(label = total_cases), vjust = -0.5, size = 3.5) +
  labs(
    title = "Total Breast Cancer Cases by Year",
    subtitle = "Aggregated Across All Age Groups",
    x = "Year",
    y = "Total Cases",
    caption = "Source: NYC Health Breast Cancer Dataset"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    plot.subtitle = element_text(size = 11, color = "gray40"),
    panel.grid.minor = element_blank()
  )
```



Reflection

Throughout this project, I gained a much deeper appreciation for the role data wrangling plays in actual analysis. Each dataset had its own set of issues — and none of them were ready to use out-of-the-box.

I learned how important it is to approach messy data with both strategy and patience. From inconsistent headers to multi-year wide formats, every issue required its own combination of `janitor`, `tidyr`, and `dplyr` tools to resolve.

Across all three datasets, visualization wasn't just the final step — it was a diagnostic tool. The plots revealed patterns, gaps, and outliers that would be missed in raw tables alone. By reshaping the data into long format and visualizing it with `ggplot2`, I transformed messy data into clear narratives. Whether it was maternal health trends, gender-based service usage, or age-driven cancer incidence, each plot was chosen to maximize interpretability and real-world relevance.